

12770 – Predictive Thermal Comfort Management for Dynamically Occupied Spaces

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The problem we are solving

Problem Statement:

Under low-cost, single-point sensor conditions, can historical temperature and humidity time-series characteristics be used to predict thermal discomfort events in semi-public spaces (such as study rooms) where personnel density dynamically changes?

Why it matters:

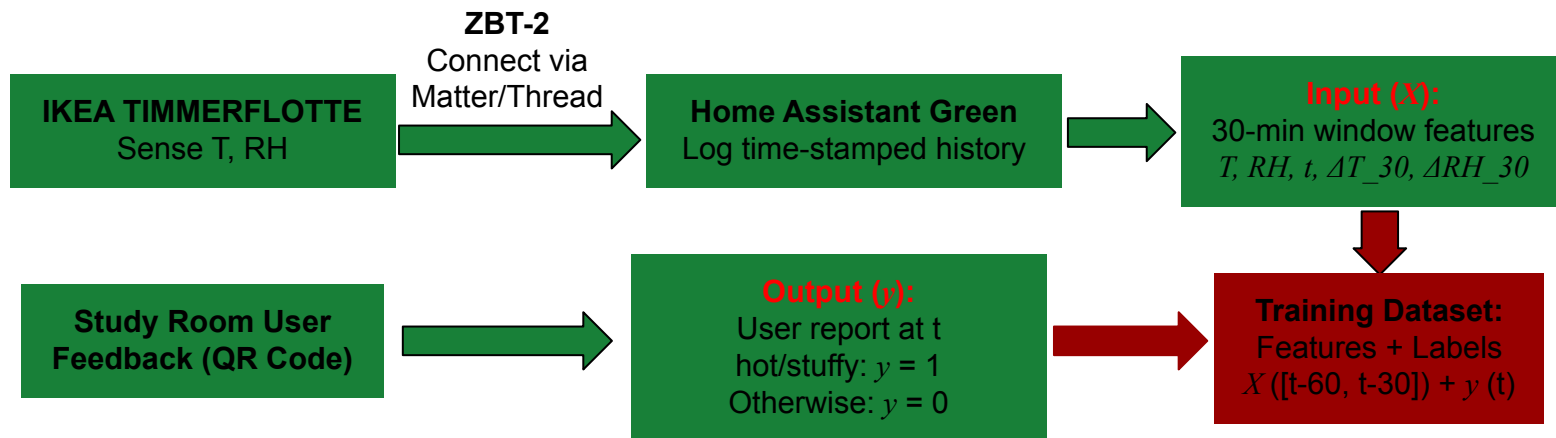
- Discomfort from heat affects concentration and learning efficiency.
- Existing conventional solutions (manual adjustments) are slow to respond, inconsistent, and non-quantifiable.

The Sense-Plan-Act Gap:

	Current Situation	Problem
Sense	Human Perception of Heat/Cold	Lag, subjective
Plan	Make decision based on experience	No prediction, random
Act	Manual adjustment	

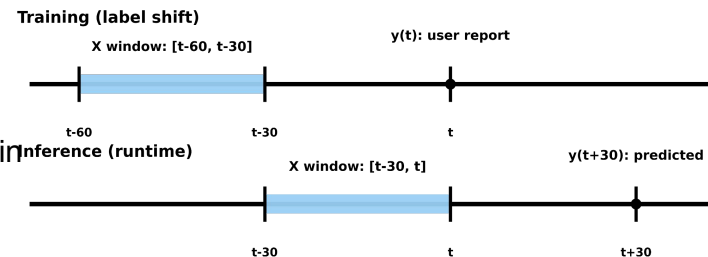
Objective: We plan to develop a low-cost system (combining low-cost sensors with machine learning technology) that can predict rising temperatures in study rooms before they reach uncomfortable levels—that is, before people even notice—and automatically propose corrective measures

Proposed Solution



Model:

- **Train** Logistic Regression / XGBoost
- **Model Output:** $p = P(y(t+30) = 1 \mid X([t-30, t]))$ – discomfort risk in next 30 min
- **Plan:** Map $p \rightarrow$ Low / Mid / High



Action:

- If **High risk** $\rightarrow$ Push alert (HA Voice announcement)
- Standard guidance: **AC** strong airflow for 10 mins + avoid frequent **window** open/close

System Validation & Evaluation Framework

1. Two-Week A/B Testing

Week 1 (Baseline): Logging data only; no system intervention.

Week 2 (Intervention): Active control + proactive AI prompts.

Metric: Reduction in manual adjustments & discomfort duration.

3. Ground Truth Calibration

Sampling: Subjective feedback 3 times daily (Morning/Noon/Night).

Purpose: Verify sensor accuracy vs. human perception

2. Offline Model Validation

Method: Time-series split (Train: first 10 days / Test: last 4 days).

Task: Predict discomfort events 30 minutes in advance.

Goal: High precision and recall in environmental forecasting.

4. Energy Efficiency Analysis

Monitoring: Real-time power tracking via smart plugs.

Metric: Energy saved per "Comfort Hour."

Goal: Optimize HVAC cycles without sacrificing comfort.

Excepted Challenges

Hardware

- The placement of sensor may affects data quality.
- Single-point sensors have limited representativeness.

Data Collection

- Collect data consistently over a continuous period (e.g., 2 weeks)
- The dataset size may be small, which could affect the model's generalization ability.

Modeling

- Risk of Overfitting in Small Datasets
- On-premises deployment for real-time reasoning

We are developing optimization strategies for each challenge and expect to manage these challenges within the scope of the project.